

The Hippocampus

The hippocampus in memory

Computational Neuroscience

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mrule

Learning outcomes:

- ▶ What happened to Henry Molaison, and what did we learn from him?
- ▶ Describe the theory of long-term memory consolidation and the putative roles of sleep, the hippocampus, and cortex.
- ▶ Be able to name and label all parts of the hippocampus involved in the ***trisynaptic loop***, and describe the role that each area plays in memory encoding.
- ▶ Dentate gyrus
 - Identify location of dentate gyrus within hippocampus
 - Identify that dentate gyrus (DG) is believed to play a role in pattern separation for memory encoding
 - Describe the role of k -winner-take-all sparsity in separating similar patterns

Hippocampus

3-layer archicortex, deep inside temporal lobe

You need it to **form new memories** and **navigate the world**



Henry Molaison

Bilateral removal of hippocampus age 27

Epilepsy controlled, but:

- ▶ Retrograde amnesia
- ▶ **Could not** form new episodic memories
- ▶ **Could** improve on motor tasks with practice



Portrait of Henry Molaison, or H.M., 1953

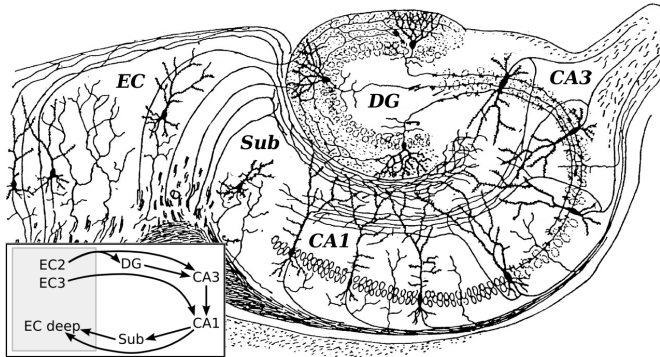
Hippocampus and memory

New episodic memories are encoded in hippocampus during the day.

During subsequent sleep, hippocampus replays the neural activity encoding the memory and triggers learning in the cortex.

Over time, cortex learns the memories, and they become hippocampus-independent.

Trisynaptic loop



Modified from Cajal (1911).

Entorhinal Cortex (EC)

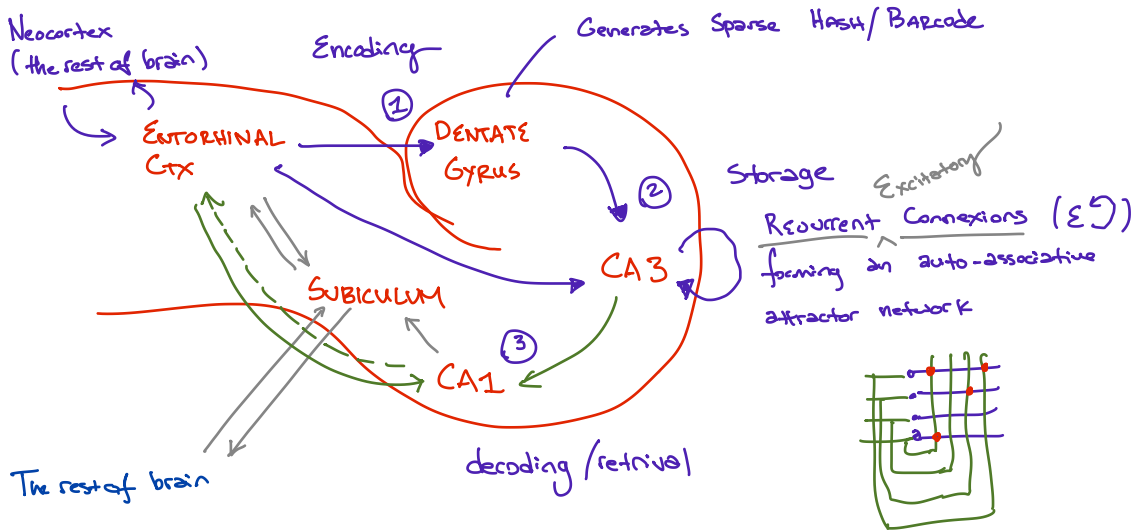
Dentate Gyrus (DG)

Cornu Ammonis field 3 (CA3)

Cornu Ammonis field 1 (CA1)

Subiculum (Sub)

draw a nice hippocampus and trisynaptic loop diagram from memory



fast Hebbian Learning

$$\Delta w_{ij} \propto y_j y_i$$

Trisynaptic loop



CA3 the **only** subregion with substantial recurrent excitatory connectivity

- ▶ recurrent **attractor network**
- ▶ pattern completion

This diagram was bad

Formally, the actual "Loop" has 4-5 synapses if you want it to be a closed loop & depending on if you stop at Sub. between CA1 & EC (depends on pathway of interest, both happen)

(Thankfully "which synapses make up trisynaptic loop" isn't explicitly an exam question.)

Entorhinal Cortex (superficial) EC_{2/3}:

- Input

Dentate Gyrus (DG): ~

- Encode distinct (albeit similar) memories differently
- **Pattern Separation**

CA3: Recurrent attractor dynamics, ✓

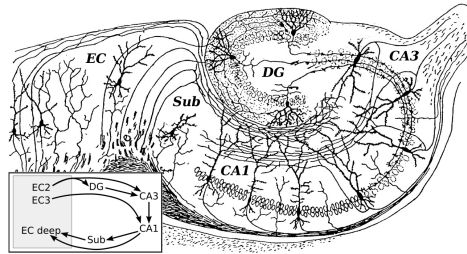
- Store patterns encountered during waking experience in recurrent weights
- **Pattern Storage**
- **Pattern Completion**

CA1: Decode

- Translate CA3 output into usable code

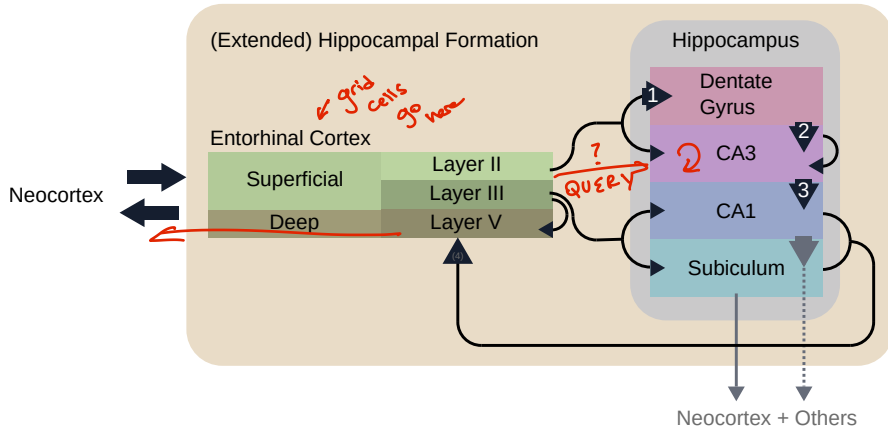
Subiculum → Entorhinal Cortex (deep) EC₄:

- Output



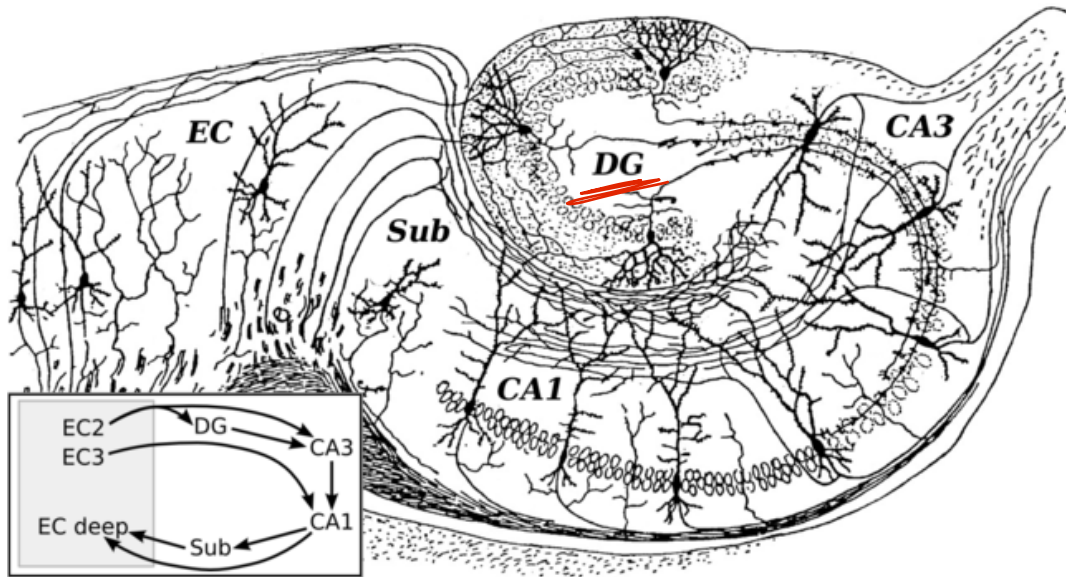
Modified from Cajal (1911).

Hippocampal formation circuit (with 3 synapses of trisynaptic loop labelled)



Modified from A. Boja

Dentate Gyrus



Cajal c. 1900

Label: path integrator, head-direction tracker, pattern separator, pattern completer

GRANULE CELLS IN DENTATE GYRUS MAKE PATTERNS SPARSE, DECORRELATED

O'Reilly & McClelland (1994):

- ▶ Distinct episodic memories → distinct CA3 attractor patterns

How can we get similar experiences → different CA3 patterns?

Encode:

DG makes pseudo-random pattern for a given EC state

CA3 responds to DG + EC_{2/3} input, and remembers the resulting activity

(CA1 updates CA3 → EC translation, if needed)

Recall:

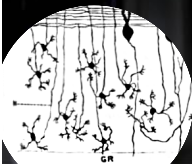
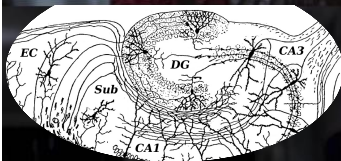
Cue from EC partly activates CA3 engramme ← Content addressable!

CA3 completes pattern

(CA1 translates reinstated memory to EC layer 4)

LEARNING PROBLEM INVOLVING
CLASSIFYING OR DISTINGUISHING
FEATURE VECTORS x

My lord, is that... **LINEARLY SEPARABLE?**

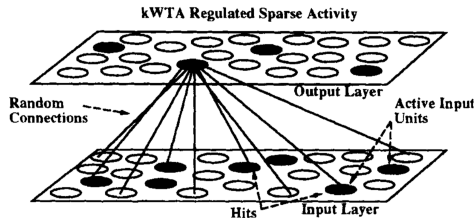
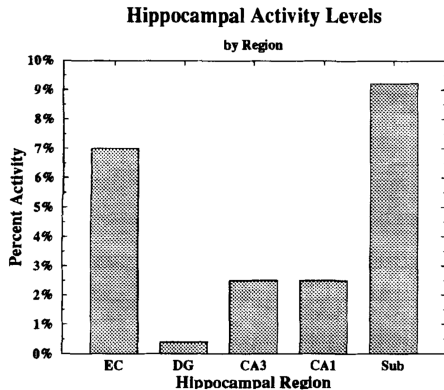


GRANULE CELLS OF
THE CEREBELLAR CORTEX
AND DENTATE GYRUS OF
THE HIPPOCAMPUS

I will make it **LINEARLY SEPARABLE**

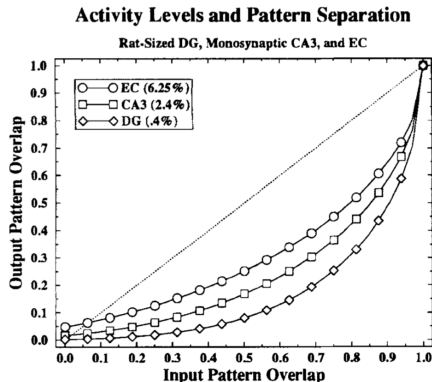
& in
case of DG
Specifically:
decorrelated /
sparse /
orthogonal
to reduce overlap
to reliably
store similar
memories as
distinct.

Random connections + k-winner-take all sparse activity = decorrelation, separation



What is the pattern overlap at the output layer as a function of the overlap in the input layer?

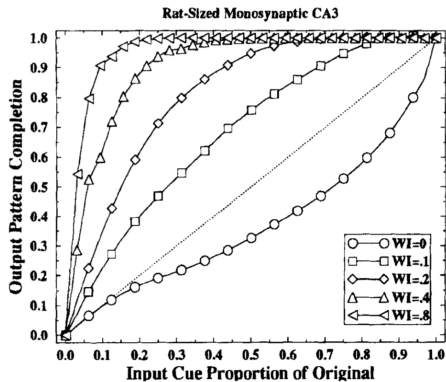
Random connections + k-winner-take all sparse activity = decorrelation, separation



A sparse, random feedforward network generically does pattern separation.

If there are k (random) spikes (1) out of N cells,

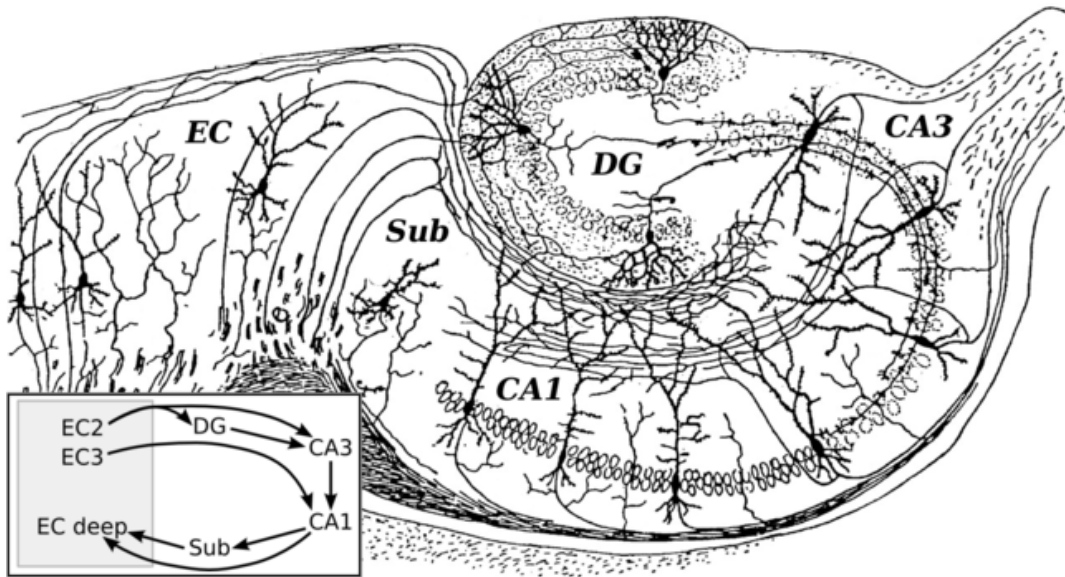
Random connections + k-winner-take all sparse activity = decorrelation, separation



Strengthening activated feedforward synapses can lead to pattern completion.

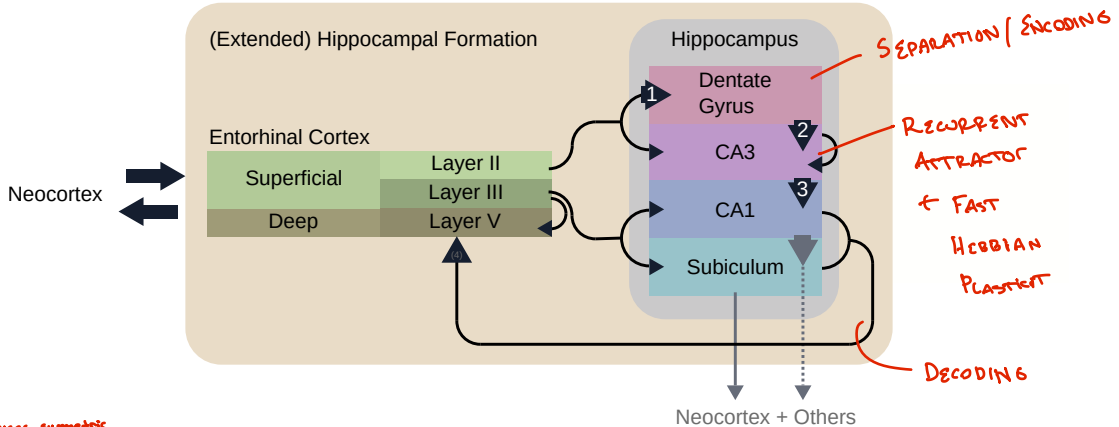
This can be enhanced by recurrent attractor dynamics in CA3.

In summary...



Cajal c. 1900

Label: path integrator, head-direction tracker, pattern separator, pattern completer



CA3 uses symmetric STDP, by the way



Modified from A. Boja

Memorize this slide for the exam.

DG handles pattern Separation: (associative hashing)

- ▶ EC → DG (perforant path)
- ▶ DG granule cells create a predictable, pseudorandom expansion of EC state → CA3
- ▶ Similar experiences → distinct memories via sparse, k -winner-take-all output

EC provides the Key/Query:

↔ communicates to/from neocortex

- ▶ EC → CA3
- ▶ EC sends to CA3 information about the current experience

CA3 stores memories in an attractor net

- ▶ CA3 → CA3
- ▶ co-active CA3 neurons strengthen the weights between them
- ▶ Sparse, separated representations easy to remember, hard to interpret

CA1 learns memory identity mapping

- ▶ CA3 → CA1 ← EC

end

